

ALLIANCE KCSE 2025 PREDICTION



POSSIBLE KCSE EXAM



CHEMISTRY

233/2

PAPER 2

(THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- a) Write your name and index number in the spaces provided
- b) Sign and write the date of examination in the spaces provided
- c) Answer ALL questions in the spaces provided
- d) Mathematical table and electronic calculators may be used.
- e) ALL working MUST be shown clearly where necessary
- f) *Candidates should check the question paper to ensure that all pages are printed as indicated and no questions are missing*

FOR EXAMINER'S USE ONLY

QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORES
1 – 6	80	

Answer ALL questions in the spaces provided

1. I. 3 g of granules of carbonate of metal M of formula MCO_3 were mixed with excess 0.5M dilute hydrochloric acid. The mass of the reaction vessel and its contents were recorded at various times. The total loss in the mass of the reaction vessel and its contents was calculated and recorded.

Time(min)	0	0.5	1	1.5	2	2.5	3	3.5
total loss in mass(Kg)	0	0.008	0.37	0.90	1.19	1.28	1.32	1.32

- a) Why is there continuous loss of mass of the reaction mixture? (1mark)

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- b) i) State two ways by which the reaction could be made more rapid. (1mark)

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- ii) On the same grid sketch, graphs of total loss in mass (g) against (mins) before and after the changes in b(i) above. (1mark)

- c) i) Write an equation for the reaction that takes place. (1mark)

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- ii) Given the experiment was carried out at room temperature, calculate the number of moles of carbon (iv) oxide gas produced. (C=12,O=16) (1mark)

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- iii) Determine the relative molecular mass of metal M. (2marks)

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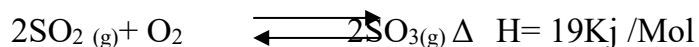
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II During the manufacture of Sulphuric (VI) acid by contact process, Sulphuric (IV) Oxide is converted to Sulphur (VI) Oxide in the presence of a catalyst according to the equation below



a) Write a formula of the catalyst used **(1mark)**

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b) What would be the effect on the production of Sulphur (VI) Oxide of the following?

i) Increasing the temperature ;Explain **(1mark)**

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ii) Reducing the pressure? Explain **(1mark)**

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c) Write equations showing how Sulphuric (VI) acid is obtained from Sulphuric (VI) Oxide

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2 a) The table below shows information about chlorides of some period 3 elements

Chloride	NaCl	MgCl ₂	AlCl ₃	SiCl ₄	PCl ₅
Melting Point (°C)	801	712	Sublimes at 183	-70	-90

i) State the nature of the solution formed when the following chlorides are dissolved in water.

Explain

(2marks)

NaCl.....

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AlCl₃.....

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ii) Explain why AlCl₃ has much lower melting point than Mg Cl₂ although both

Al and Mg are metals

(2marks)

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iii) Draw (.) and cross(x) diagram to show bonding in:

MgCl₂

(1mark)

SiCl₄

(1mark)

iv) Identify the type of bonding in each case

(1mark)

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b) 10cm³ of water was placed in a test tube and the initial temperature of the water recorded. A spatula of Si Cl₄ was added to the water and the final temperature recorded when the entire solid had dissolved.

i) Explain the observations made.

(2marks)

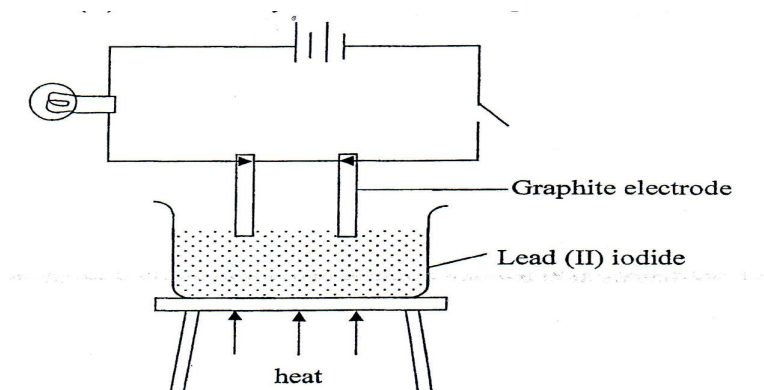
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ii) Write an equation for the reaction that occurred

(1mark)

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3. The diagram below shows a setup which was used to investigate the effect of electricity on solid and molten lead(II) iodide. Study it and answer the questions that follow.



a) i) The bulb only lit after the Lead (II) iodide had melted .Explain

(1mark)

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ii) State and explain the observation at the anode when the circuit is completed.

(1mark)

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iii) What precaution should be taken when carrying out this experiment

(1mark)

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iv) Write the equation for the reaction taking place as the cathode (1mark)

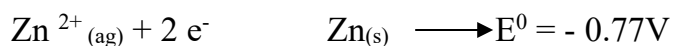
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v) On the diagram; indicate the movement of ions in the electrolyte. (1mark)

vi) When a current was passed through 23.6g of molten Lead (II) iodide for 3 mins 5 secs 19.1g of the electrolyte remained. Calculate the amount of current passed. (3marks)

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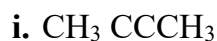
b) Given the following standard electrode potentials;



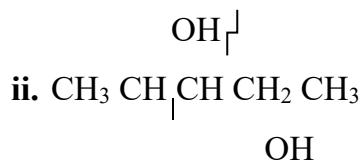
Explain whether it's advisable to store a solution of Zinc sulphate in a copper can. (2marks)

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4. a) Name the following organic Compounds (3marks)



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b) i) Define isomerism (1mark)

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ii) Draw and name one branching isomer of C₅H₈ (2marks)

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iii) To which homologous series does C₅H₈ belong? State the general formula of the members of the series (1mark)

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c) Identify two reagents that can be used to prepare 3 - Bromobutan – 2-ol in the laboratory (1mark)

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d) In an experiment, ethanol was warmed with acidified potassium manganate (VI). State and explain the observation made at the end of the experiment (2marks)

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5. In an experiment to investigate the enthalpy of displacement of copper by iron, 25.0 cm³ of Copper (II) sulphate solutions was placed in plastic beaker. The temperature of the solution was recorded every half minute of 1.5 minutes. At exactly two minutes excess iron powder was added and the temperature of the mixture recorded every half minute for an additional three minutes. The results were recorded in the table below

Time (min)	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
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Temp $^{\circ}\text{C}$	18.0	18.0	18.0	18.0	X	31.0	29.0	28.5	25.0	23.0	21.0
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- a) Plot a graph of Temperature ($^{\circ}\text{C}$) against time (mins) (3marks)

Graph paper provided

- b) From the graph, determine the highest temperature, hence the temperature change. (2marks)

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- c) Write the ionic equation for the reaction. (1mark)

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- d) Calculate the amount of heat given out during the reaction

($c = 4.2 \text{ kJ/Kg/K}$, density of solution = 1 g/cm^3) (2marks)

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- e) Given the molar heat for the displacement in the above reaction is -151 kJ/Mol , calculate:

- i) The moles of Copper (II) sulphate reacted (1mark)

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- ii. The concentration of the copper (II) sulphate solution in g/litre (2marks)

($\text{Cu} = 63.5, \text{S} = 32, \text{O} = 16$)

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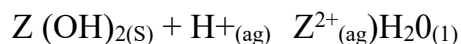
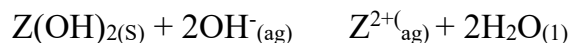
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6. a) A compound whose formula is $Z(OH)_2$ reacts as shown by the equations below



i) State the property exhibited by the compound in the reactions (1mark)

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ii) Name two elements whose hydroxides exhibit the above property. (1mark)

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b) The table below shows tests carried out on a sample of water.

Test	Results
Additional of sodium hydroxide drop wise till in excess	White precipitate which is insoluble in excess
2. Addition of sodium Sulphate Solution	White precipitate
3. Addition of Lead (II) nitrate solution .Then warming	White precipitate , which dissolves on warming.

i) Identify the anion present in the water (1mark)

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ii) Write an ionic equation for the reaction in step 1 (1mark)

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iii) Given two methods that can be used to soften the water sample (2marks)

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c) The following tests were carried out on Solid M.

i) Solid M was added to water and the mixture stirred, a colorless solution was formed

ii) Barium nitrate was added to a portion of the colourless solution, a white precipitate was formed which dissolved in dilute Nitric (V) acid to form a colourless solution.

iii) Orange acidified potassium dichromate (VI) changed to green when added to a portion of the solution

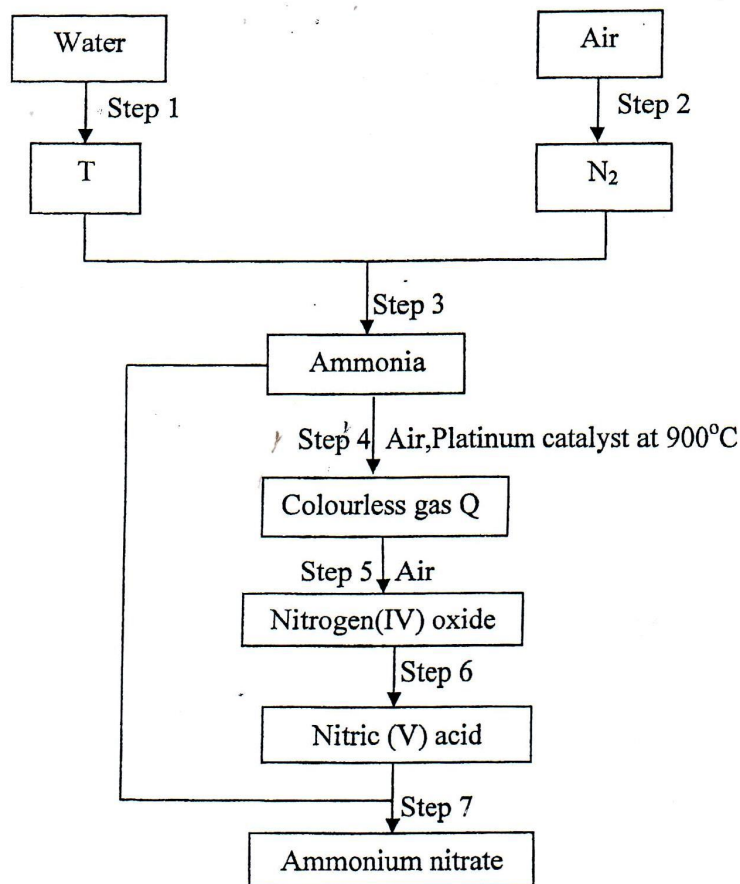
iv) When Chlorine gas was bubbled into a portion of the colourless solution followed by addition of acidified Barium nitrate solution, a white precipitate was formed.

Name and write the formula of white precipitate formed in:-9

Step II..... (1mark)

Step IV..... (1mark)

d) Study the flow chart below and answer the questions that follow.



i) Name element t (1 ½ marks)

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ii) Identify the process in (1 ½ marks)

Step 1.....

Step 2.....

Step 4.....

iii) Identify colourless gas

Q..... .(1½ marks)

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